Solution 1 Brownian Bridge

Problem 1 — We set $W_t = B_t - tB_1$ for every $t \in [0,1]$.

Problem 1.1 — Show that $(W_t)_{t\in[0,1]}$ is a centered Gaussian process and give its covariance function.

Proof. We have that W_t is a linear combination of Gaussian processes, and so is itself Gaussian. It is centered since B_t is centered. The covariance function is given by

$$\begin{aligned}\operatorname{cov}(W_s, W_t) &= \operatorname{cov}(B_s - sB_1, B_t - tB_1)\\ &= \operatorname{cov}(B_s, B_t) - s\operatorname{cov}(B_1, B_t) - t\operatorname{cov}(B_s, B_1) + st\operatorname{cov}(B_1, B_1)\\ &= s \wedge t - st,\end{aligned}$$

where we have used the fact that B_t is a Brownian motion, so that $\operatorname{cov}(B_s, B_t) = s \wedge t$. □

Problem 1.2 — Let $0 < t_1 < t_2 < \cdots < t_p < 1$. Show that the law of $(W_{t_1}, W_{t_2}, \ldots, W_{t_p})$ has density

$$g(x_1, x_2, \ldots, x_p) = \sqrt{2\pi} p_{t_1}(x_1) p_{t_2-t_1}(x_2 - x_1) \cdots p_{t_p - t_{p-1}}(x_p - x_{p-1}) p_{1-t_p}(-x_p),$$

where $p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t}$, and explain why the law of $(W_{t_1}, W_{t_2}, \ldots, W_{t_p})$ can be interpreted as the conditional law of $(B_{t_1}, B_{t_2}, \ldots, B_{t_p})$ knowing that $B_1 = 0$.

Proof. Observe that $B_t - tB_1$ is independent of B_1 for all $t \in [0,1]$. Indeed,

$$\operatorname{cov}(B_t - tB_1, B_1) = \operatorname{cov}(B_t, B_1) - t\operatorname{cov}(B_1, B_1) = t - t = 0.$$

Thus, the density functions of $(W_{t_1}, \ldots, W_{t_p}, B_1)$ (denoted f) and B_1 (denoted p_1) will satisfy

$$f(x_1, \ldots, x_p, y) = g(x_1, \ldots, x_p) p_1(y).$$

Idea — Everything here is a GRV with support on its domain (I can draw any small ball in $\mathbb{R}^{p+1}$ and there is some nonzero chance that $(W_{t_1}, \ldots, W_{t_p}, B_1)$ will lie in that ball, etc), so all of these densities exist.

However, it's pretty easy to compute f. Let h be the density function of $(B_{t_1}, \ldots, B_{t_p}, B_1)$.

$$f(x_1, \ldots, x_p, y) = h(x_1 + t_1 y, \ldots, x_p + t_p y, y).$$

Idea (The Jacobian is one) — This linear transformation is the matrix

$$\begin{pmatrix} 1 & t_1 & 0 & \cdots & 0 \\ 0 & 1 & t_2 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix},$$

> which has determinant one. So when we use change of variables, there's no Jacobian to worry about.

We know h and p_1 though. Let $t_0 = 0$, $x_0 = 0$, $t_{p+1} = 1$, and $x_{p+1} = 0$.

$$h(x_1 + t_1 y, \ldots, x_p + t_p y, y) = \prod_{i=1}^{p+1} \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} e^{-((x_i+t_i y)-(x_{i-1}+t_{i-1}y))^2/2(t_i-t_{i-1})}$$

Just stare at the exponents for a while. We have

$$\begin{aligned}\sum_{i=1}^{p+1} \frac{((x_i + t_i y) - (x_{i-1} + t_{i-1}y))^2}{2(t_i - t_{i-1})} &= \sum_{i=1}^{p+1} \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} \\ &\quad + 2y \sum_{i=1}^{p+1} \frac{(x_i - x_{i-1})(t_i - t_{i-1})}{2(t_i - t_{i-1})} + y^2 \sum_{i=1}^{p+1} \frac{(t_i - t_{i-1})^2}{2(t_i - t_{i-1})} \\ &= \sum_{i=1}^{p+1} \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} + \frac{y^2}{2}.\end{aligned}$$

Thus,

$$h(x_1 + t_1 y, \ldots, x_p + t_p y, y) = \left(\prod_{i=1}^{p+1} \underbrace{\frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} e^{-(x_i - x_{i-1})^2/2(t_i - t_{i-1})}}_{p_{t_i - t_{i-1}}(x_i - x_{i-1})} \right) p_1(y).$$

and

$$g(x_1, \ldots, x_p) = \sqrt{2\pi} p_{t_1}(x_1) p_{t_2 - t_1}(x_2 - x_1) \cdots p_{t_p - t_{p-1}}(x_p - x_{p-1}) p_{1-t_p}(-x_p).$$

as desired. This can be interpreted as a conditional law, because the law of $B_{t_1}, \ldots, B_{t_p}, B_1$ is

$$p_{t_1}(x_1) p_{t_2 - t_1}(x_2 - x_1) \cdots p_{t_p - t_{p-1}}(x_p - x_{p-1}) p_{1-t_p}(x_{p+1} - x_p)$$

and this is that, evaluated at $x_{p+1} = 0$, divided by $p_1(0) = \frac{1}{\sqrt{2\pi}}$. □

> **Problem 1.3 —** Verify that the two processes $(W_t)_{t\in[0,1]}$ and $(W_{1-t})_{t\in[0,1]}$ have the same distribution (similarly as in the definition of Wiener measure, this law is a probability measure on the space of all continuous functions from $[0, 1]$ into $\mathbb{R}$).

Proof. The density we just showed is manifestly time-reversal-invariant. Of course, prescribing this density is equivalent to specifying the law on all finite-dimensional marginals, i.e. the probability measure over any cylinder set consisting of functions which pass through Borel sets $A_1, \ldots A_n$ at times $t_1 < \cdots < t_n$. These cylinder sets form a generating π-system for the product σ-algebra $\mathbb{R}^{\otimes[0,1]}$.

The σ-algebra on continuous functions $C([0, 1], \mathbb{R})$ is just the intersection of the product σ-algebra with the set of continuous functions, thus the images of the cylinder sets remain a π-system. Explicitly, these images look like the set of all continuous functions which pass through $A_1, \ldots, A_n$ at times $t_1 < \cdots < t_n$.

In any case, this just means that the collection of finite-dimensional marginals uniquely specifies a law on the entire σ-algebra over $C([0, 1], \mathbb{R})$, if such a law exist. Thus the law of $(W_t)_{t\in[0,1]}$ is the same as that of $(W_{1-t})_{t\in[0,1]}$. □

Solution 2 Time Reversal for Brownian Motion

Problem 2 — We set $B'_t = B_1 - B_{1-t}$ for every $t \in [0,1]$. Show that the two processes $(B_t)_{t\in[0,1]}$ and $(B'_t)_{t\in[0,1]}$ have the same law (as in the definition of Wiener measure, this law is a probability measure on the space of all continuous functions from $[0,1]$ into $\mathbb{R}$).

Idea (Hint) **—** Note that the choice of end time $t = 1$ is not special: it will follow by essentially the same argument that for any fixed $t \geq 0$, the process $(B_t - B_{t-s} : 0 \leq s \leq t)$ is a standard Brownian motion on the time interval $[0,t]$.

We compute the covariance function of $(B'_t)_{t\in[0,1]}$:

$$\begin{aligned}
\operatorname{cov}(B'_s, B'_t) &= \operatorname{cov}(B_1 - B_{1-s}, B_1 - B_{1-t}) \\
&= \operatorname{cov}(B_1, B_1) - \operatorname{cov}(B_1, B_{1-t}) - \operatorname{cov}(B_{1-s}, B_1) + \operatorname{cov}(B_{1-s}, B_{1-t}) \\
&= 1 - (1-t) - (1-s) + (1-s) \wedge (1-t) \\
&= s + t - s \vee t \\
&= s \wedge t.
\end{aligned}$$

Thus, $(B_t)_{t\in[0,1]}$ and $(B'_t)_{t\in[0,1]}$ are both centered Gaussian processes with the same covariance matrices. Also, B'_t is obviously path-continuous. Thus, both processes induce the same Wiener measure. (The Weiner measure is unique, as stated in the textbook; being a Gaussian process automatically yields all finite-dimensional distributions, and these form a generating π-system; see the previous problem).

Solution 3 Reflected Brownian Motion

Problem 3 — Let B_t be a standard Brownian motion, $S_t = \sup\{B_s : 0 \le s \le t\}$ its running maximum, and $Y_t = S_t - B_t \ge 0$. In this exercise you will show that the processes $(Y_t)_{t\ge 0}$ and $(|B_t|)_{t\ge 0}$ are equal in law. Note: be careful that $(S_t)_{t\ge 0}$ is in fact not a Markov process, although it will follow from this exercise that $(Y_t)_{t\ge 0}$ is.

Problem 3.1 — First argue that to prove the claim, it suffices to show that for any fixed $s, t \ge 0$, conditional on $\mathscr{F}_s$ the random variable Y_{s+t} is equidistributed as $\left|Y_s + \bar{B}_t\right|$ where $\bar{B}$ denotes a Brownian motion independent of $\mathscr{F}_s$.

What it means for $(Y_t)_{t\ge 0}$ and $(|B_t|)_{t\ge 0}$ to have the same law is that for all $t_1, t_2, \ldots, t_n \ge 0$, the joint distribution of $(Y_{t_1}, Y_{t_2}, \ldots, Y_{t_n})$ is the same as the joint distribution of $(|B_{t_1}|, |B_{t_2}|, \ldots, |B_{t_n}|)$. The key is to break this up into conditional distributions, and then use the fact that the law of $(B_t)_{t\ge 0}$ is invariant under time reversal.

Start with the expression

$$\mathbb{P}\left(Y_{s+t} \in A \mid \mathscr{F}_s\right) = \mathbb{P}\left(\left|Y_s + \bar{B}_t\right| \in A \mid \mathscr{F}_s\right), \quad a \ge 0, \qquad A \in \mathscr{B}(\mathbb{R}).$$

We will now show by induction that

$$\mathbb{P}(\bigcap_i (Y_{t_i} \in A_i)) = \mathbb{P}(\bigcap_i (|B_{t_i}| \in A_i))$$

for all $t_1, t_2, \ldots, t_n \ge 0$ and $A_1, A_2, \ldots, A_n \in \mathscr{B}(\mathbb{R})$. (Note that this equivalent to $(Y_{t_1}, \ldots, Y_{t_n})$ and $(|B_{t_1}| \ldots, |B_{t_n}|)$ having the same distribution on $\mathbb{R}^n$). The base case is $n = 0$ which is trivial.

Then, as the inductive step,

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{i\le n}(Y_{t_i} \in A_i)\right) &= \mathbb{P}\left(\left(\left|Y_{t_{n-1}} + \bar{B}_{t_n - t_{n-1}}\right| \in A_n\right) \cap \bigcap_{j<n}(Y_{t_j} \in A_j)\right) \\
&= \mathbb{P}\left(\left(\left|\left|B_{t_{n-1}}\right| + (B_{t_n} - B_{t_{n-1}})\right| \in A_n\right) \cap \bigcap_{j<n}(\left|B_{t_j}\right| \in A_j)\right) \\
&= \mathbb{P}\left(\bigcap_{j\le n}(\left|B_{t_j}\right| \in A_j)\right).
\end{aligned}$$

The first line is because conditioned on $\bigcap_{j<n}(Y_{t_j} \in A_j) \in \mathscr{F}_{t_{n-1}}$, $\left|Y_{t_{n-1}} + \bar{B}_{t_n - t_{n-1}}\right|$ has the same law as Y_{t_n} as postulated in the problem statement. The second line is because by the inductive hypothesis, $(\left|B_{t_j}\right|)_{j\le n-1}$ and $(Y_{t_j})_{j\le n-1}$ have the same law – and $\bar{B}_{t_n - t_{n-1}}$ is just a $N(0, t_n - t_{n-1})$-distributed RV independent of $\mathscr{F}_s$, just like $B_{t_n} - B_{t_{n-1}}$.

The last line is because conditioned on $B_{t_{n-1}} \ge 0$, $|B_{t_n - 1}| + (B_{t_n} - B_{t_{n-1}}) = B_{t_n}$, but even if $B_{t_{n-1}} < 0$, $(B_{t_n} - B_{t_{n-1}})$ is still just a $N(0, t_n - t_{n-1})$-distributed RV independent of $\mathscr{F}_s$, and in particular can be replaced by $-(B_{t_n} - B_{t_{n-1}})$ yielding $-B_{t_n}$ as desired.

The inductive hypothesis is precisely the claim that $(Y_t)_{t\ge 0}$ and $(|B_t|)_{t\ge 0}$ have the same law (we have verified it over all finite-dimensional cylinder sets, which form a π-system generating the product σ-algebra). Thus, we have shown that the two processes have the same law.

Problem 3.2 — For $u \geq 0$ let $\hat{B}_u \equiv B_{s+u} - B_s$ and $\hat{S}_u = \sup\{\hat{B}_w : 0 \leq w \leq u\}$. Show that $Y_{s+t} = \max\{Y_s, \hat{S}_t\} - \hat{B}_t$.

This is just true by explicit computation:

$$\begin{aligned}
\max\{Y_s, \hat{S}_t\} - \hat{B}_t &= \max\{S_s - B_s, \sup\{\hat{B}_w : 0 \leq w \leq t\}\} - \hat{B}_t \\
&= \max\{S_s - B_s, \sup\{B_{s+w} - B_s : 0 \leq w \leq t\}\} - (B_{s+t} - B_s) \\
&= \max\{S_s, \sup\{B_{s+w} : 0 \leq w \leq t\}\} - B_{s+t} \\
&= S_{s+t} - B_{s+t} \\
&= Y_{s+t}.
\end{aligned}$$

Problem 3.3 — Argue that for any fixed $y \geq 0$, $\max\{y, \hat{S}_t\} - \hat{B}_t$ has the same law as $\left|y + \hat{B}_t\right|$.

Idea (Hint) — A hint is to first decompose

$$\mathbb{P}\left(\max\{y, \hat{S}_t\} - \hat{B}_t > a\right) = \mathbb{P}\left(y - \hat{B}_t > a\right) + \mathbb{P}\left(y - \hat{B}_t \leq a, \hat{S}_t - \hat{B}_t > a\right),$$

and use time reversal to simplify the last term.

We apply the hint. Let $\hat{B}'_s = \hat{B}_{t-s} - \hat{B}_t$ be the time-reversed process, and $\hat{S}'_t = \sup\{\hat{B}'_w : 0 \leq w \leq t\}$. Observe that $\hat{B}'_t = -\hat{B}_t$ and $\hat{S}'_t = \hat{S}_t - \hat{B}_t$.

$$\begin{aligned}
\mathbb{P}(y - \hat{B}_t \leq a, \hat{S}_t - \hat{B}_t > a) &= \mathbb{P}(\hat{B}'_t \leq a - y, \hat{S}'_t > a) \\
&= \mathbb{P}(\hat{B}'_t > y + a) \\
&= \mathbb{P}(y - \hat{B}'_t < -a) \\
&= \mathbb{P}(y + \hat{B}_t < -a)
\end{aligned}$$

where the second line is by the refection principle. Summing,

$$\mathbb{P}\left(\max\{y, \hat{S}_t\} - \hat{B}_t > a\right) = \mathbb{P}\left(y - \hat{B}_t > a\right) + \mathbb{P}\left(y - \hat{B}_t \leq a, \hat{S}_t - \hat{B}_t > a\right) = \mathbb{P}\left(\left|y + \hat{B}_t\right| > a\right).$$

as desired. (Note that $\mathbb{P}(y - \hat{B}_t > a) = \mathbb{P}(y + \hat{B}_t > a)$ because Brownian motion is symmetric under reflection).

Solution 4 Arcsine Law

Problem 4 — Set $T = \inf\{t \geq 0 : B_t = S_1\}$.

Problem 4.1 — Show that $T < 1$ a.s. (one may use the result of the previous exercise) and then that T is not a stopping time.

We know the Brownian motion is continuous almost surely. Continuous functions on compact domains attain their maximum, thus there exists a $t \in [0, 1]$ such that $B_t = S_1$ almost surely. It remains to show that $B_1 = S_1$ almost never. To do this, we use time reversal; let $B'_t = B_{1-t} - B_1$, and note that B'_t is a Brownian motion; then $S'_1 = \inf\{B'_t, 0 \leq t \leq 1\} = S_1 - B_1$. However, as a Brownian motion, we know that almost surely for every $\varepsilon > 0$,

$$\sup_{0 \leq t \leq \varepsilon} B'_t > 0$$

thus $S'_1 > 0$ almost surely, i.e. $S_1 > B_1$ almost surely. So we conclude.

This is not a stopping time, because it does not obey the strong Markov property. Indeed, setting $B_t^{(T)} = B_{T+t} - B_T$, we observe that $B^{(T)}$ is not a Brownian motion because it is forced to be nonpositive on a small interval $[0, \varepsilon)$! (ε is positive because $T < 1$ almost surely). Thus, T is not a stopping time.

Problem 4.2 — Verify that the three variables S_t, $S_t - B_t$, and $|B_t|$ have the same law.

From the preceding problem, we have that the processes $(S_t - B_t)_{t \geq 0}$ and $(|B_t|)_{t \geq 0}$ have the same law. From theorem 2.21 in the textbook, S_t has the same distribution as $|B_t|$.

Problem 4.3 — Show that T is distributed according to the so-called arcsine law, whose density is

$$g(t) = \frac{1}{\pi\sqrt{t(1-t)}} \mathbb{1}_{(0,1)}(t).$$

We are concerned with $\mathbb{P}(T \leq a) = \mathbb{P}(S_a = S_1)$. Consider the two processes $B_t^L = B_{a-t} - B_a$ for $0 \leq t \leq a$ and $B_t^R = B_{a+t} - B_a$ for $0 \leq t \leq 1 - a$. For each t, B_t^L lives in $\mathscr{F}_a$, and thus is independent of B_s^R. Allowing $S_t^L = \sup\{B_s^L : 0 \leq s \leq t\}$ and $S_t^R = \sup\{B_s^R : 0 \leq s \leq t\}$, we have that $S_a^L = S_a - B_a$ and $S_{1-a}^R = \sup\{B_s : a \leq s \leq t\} - B_a$. Thus, $S_a = S_1$ iff $S_a^L \geq S_{1-a}^R$.

But we know that S_a^L is independent of S_{1-a}^R, and these are distributed as $|B_a|$ and $|B_{1-a}|$, which are the absolute values of $N(0, a)$ and $N(0, 1-a)$. We can explicitly perform the integral for $\mathbb{P}(|B_a| \geq |B_{1-a}|)$:

$$\int_0^\infty \int_0^x \frac{2}{\sqrt{2\pi a}} e^{-y^2/2a} \frac{2}{\sqrt{2\pi(1-a)}} e^{-x^2/2(1-a)} \, dy \, dx = \int_0^\infty \int_0^{\sqrt{\frac{a}{1-a}}x} \frac{4}{2\pi} e^{-(x^2+y^2)/2} \, dy \, dx.$$

Expressing in radial coordinates, the problem becomes trivial:

$$\int_0^{\sin^{-1}(\sqrt{a})} \int_0^\infty \frac{4}{2\pi} e^{-r^2/2} r \, dr \, d\theta = \int_0^{\sin^{-1}(\sqrt{a})} \frac{2}{\pi} d\theta = \frac{2}{\pi} \sin^{-1}(\sqrt{a}).$$

Thus the density is

$$g(t) = \frac{d}{dt}\left(\frac{2}{\pi}\sin^{-1}(\sqrt{t})\right) = \frac{2}{\pi}\frac{1}{\sqrt{1-t}} \cdot \frac{1}{2\sqrt{t}} = \frac{1}{\pi\sqrt{t(1-t)}}$$

as desired.

Problem 4.4 — Show that the results of questions 1. and 3. remain valid if T is replaced by $L = \sup\{t \leq 1 : B_t = 0\}$.

Idea (Hint) — You may use the result of Problem 3 in your solution; it is especially useful for the last part of this exercise.

By problem 3, $L = \sup\{t \leq 1 : |B_t| = 0\}$ and $T' = \sup\{t \leq 1 : B_t = S_1\} = \sup\{T \leq 1 : S_t - B_t = 0\}$ have the same distribution. By time reversal, T' is equi-distributed as T. This shows that $L < 1$ almost surely and is distributed according to the arcsine law.

Now, L is not a stopping time because it does not obey the strong Markov property. Indeed, setting $B_t^{(L)} = B_{L+t}$, we observe that $B^{(L)}$ is not a Brownian motion, because on any small interval $(0, \varepsilon)$ it is forced to be nonzero! Thus, L is not a stopping time.

Solution 5 Law of the Iterated Logarithm

Problem 5 — The goal of the exercise is to prove that

$$\limsup_{t\to\infty} \frac{B_t}{\sqrt{2t\log\log t}} = 1 \quad \text{a.s.}$$

We set $h(t) = \sqrt{2t\log\log t}$.

Idea (Hint) — In your solution, please give a proof of the following standard Gaussian tail bound: if $Z \sim \mathcal{N}(0,1)$, then for all $x > 0$ we have

$$\left(1 - \frac{1}{x^2}\right)\exp\left(-\frac{x^2}{2}\right)\frac{1}{x\sqrt{2\pi}} \le \mathbb{P}(Z \ge x) \le \exp\left(-\frac{x^2}{2}\right)\frac{1}{x\sqrt{2\pi}}.$$

The upper bound is easy. Observe

$$x\mathbb{P}(Z \ge x) \le \frac{1}{\sqrt{2\pi}}\int_x^\infty te^{-t^2/2}dt = \exp\left(-\frac{x^2}{2}\right)\frac{1}{\sqrt{2\pi}}$$

I'll prove a strictly stronger lower bound;

$$\left(\frac{x}{x^2+1}\right)\exp\left(-\frac{x^2}{2}\right)\frac{1}{x\sqrt{2\pi}} \le \mathbb{P}(Z \ge x).$$

(This is stronger, because for all $x > 0$, $\frac{1}{x} - \frac{1}{x^3} \le \frac{x}{x^2+1} \iff x^4 - 1 \le x^4$).

To do this, we observe three things about the function $\mathbb{P}(Z \ge x) - \left(\frac{x}{x^2+1}\right)e^{-x^2/2}/(x\sqrt{2\pi})$:

- At $x = 0$, it is positive.
- As $x \to \infty$, it vanishes.
- For $x > 0$, the derivative is

$$\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\left(1 - \frac{x^4+2x^2-1}{(x^2+1)^2}\right) = -\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\frac{2}{(x^2+1)^2} < 0.$$

  Thus this function is strictly decreasing on $x > 0$.

This means this function is positive on $[0,\infty)$, as desired.

Problem 5.1 — Show that, for every $t > 0$, $\mathbb{P}(S_t > u\sqrt{t}) \sim \frac{2}{u\sqrt{2\pi}}e^{-u^2/2}$ when $u \to +\infty$.

Brownian motion is invariant under scaling; let $B_t^\lambda = \frac{1}{\lambda}B_{\lambda^2 t}$; then B_t^λ is a Brownian motion. In particular, $S_t^\lambda = \sup\{B_s^\lambda : 0 \le s \le t\} = \frac{1}{\lambda}S_{\lambda^2 t}$. Setting $\lambda = \sqrt{t}$, it suffices to show that

$$\mathbb{P}(S_1 > u) \sim \frac{2}{u\sqrt{2\pi}}e^{-u^2/2} \quad \text{as } u \to \infty.$$

We know S_1 is distributed the same as $|B_1|$, and thus the bound we showed above implies

$$\left(1 - \frac{1}{u^2}\right)\exp\left(-\frac{u^2}{2}\right)\frac{2}{u\sqrt{2\pi}} \le \mathbb{P}(S_1 > u) \le \exp\left(-\frac{u^2}{2}\right)\frac{2}{u\sqrt{2\pi}}.$$

as desired.

Problem 5.2 — Let r and c be two real numbers such that $1 < r < c^2$. From the behavior of the probabilities $\mathbb{P}(S_{r^n} > ch(r^{n-1}))$ when $n \to \infty$, infer that, a.s.,

$$\limsup_{t\to\infty} \frac{B_t}{h(t)} \leq 1.$$

Write out

$$\begin{aligned}
\mathbb{P}(S_{r^n} > ch(r^{n-1})) &= \mathbb{P}(S_{r^n} > c\sqrt{2r^{n-1}\log\log r^{n-1}}) \\
&= \mathbb{P}(S_{r^n} > r^{n/2}\underbrace{c\sqrt{2r^{-1}\log\log r^{n-1}}}_{u}) \\
&\leq \frac{2}{c\sqrt{2r^{-1}\log\log r^{n-1}}\sqrt{2\pi}} \exp\left(-\left(c\sqrt{2r^{-1}\log\log r^{n-1}}\right)^2/2\right) \\
&= \frac{2}{c\sqrt{2r^{-1}\log\log r^{n-1}}\sqrt{2\pi}} \exp\left(-c^2r^{-1}\log\log r^{n-1}\right) \\
&= \frac{2}{c\sqrt{2r^{-1}\log\log r^{n-1}}\sqrt{2\pi}} \left((n-1)\log r\right)^{-c^2r^{-1}} \\
&\overset{n\to\infty}{\leq} K\left((n-1)\log r\right)^{-c^2r^{-1}}
\end{aligned}$$

The final inequality is true in the $n \to \infty$ limit, where K is some constant that absorbs the polylogarithmic garbage (which is decreasing with n anyways). The bound scales as $n^{-c^2/r}$ where $c^2/r > 1$. Thus, the sum of these probabilities across $1 \leq n$ is finite. By Borel-Cantelli, the event that $S_{r^n} > ch(r^{n-1})$ infinitely often has probability 0.

Now, suppose for sake of contradiction that $\limsup_{t\to\infty} B_t/h(t) > 1$ with nonzero probability. Then, I can find a $c^2 > 1$ such that $\limsup_{t\to\infty} B_t/h(t) > c$ with nonzero probability. Then, $S_t > B_t$, thus $\limsup_{t\to\infty} S_t/h(t) > c$ with nonzero probability. Pick any $1 < r < c^2$. For any t, I can find an n such that $r^{n-1} \leq t < r^n$. Observe both S_t and $h(t)$ are nondecreasing, and we find that $S_{r^n}/h(r^{n-1}) > c$ i.o. with nonzero probability. But we just showed this occurs with probability zero.

Problem 5.3 — Show that a.s. there are infinitely many values of n such that

$$B_{r^n} - B_{r^{n-1}} \geq \sqrt{\frac{r-1}{r}}h(r^n).$$

Conclude that the statement given at the beginning of the exercise holds.

Let E_n be the event that $B_{r^n} - B_{r^{n-1}} \geq \sqrt{\frac{r-1}{r}}h(r^n)$; these are independent events. $B_{r^n} - B_{r^{n-1}}$ is a centered Gaussian random variable with variance $t = r^n - r^{n-1}$. Compute

$$\sqrt{\frac{r-1}{r}}h(r^n) = \sqrt{\frac{r-1}{r}}\sqrt{2r^n\log\log r^n} = \sqrt{t}\underbrace{\sqrt{2\log\log r^n}}_{u}$$

By earlier bounds we know that

$$\mathbb{P}(B_{r^n} - B_{r^{n-1}} > u\sqrt{t}) \sim \frac{2}{u\sqrt{2\pi}}\frac{1}{n\log r}$$

The sum of these probabilities is unbounded.

Idea (Really?) — This is well-known, but let's do it anyways. We are looking at constants times $\sum_{n\geq 1} \frac{1}{n\sqrt{\log n}}$. By the Cauchy condensation test, this sum converges if and only if this guy does:

$$\sum_{n\geq 1} 2^n \frac{1}{2^n\sqrt{\log 2^n}} = \sum_{n\geq 1} \frac{1}{\sqrt{n\log 2}} = \Theta\left(\frac{1}{\sqrt{\log n}}\right)$$

But this guy explodes.

(Anyways, the mnemonic is that $\sum \frac{1}{n\log n} \sim \log\log n$).

Thus, by the other half of Borel-Cantelli, the event that $B_{r^n} - B_{r^{n-1}} \geq \sqrt{\frac{r-1}{r}} h(r^n)$ infinitely often has probability 1. This concludes the proof.

Now, by the results above, we know that $\limsup \frac{B_t}{h(t)} \leq 1$ almost surely. By symmetry, $\limsup \frac{|B_t|}{h(t)} \leq 1$ almost surely. In particular, with probability one, $\frac{|B_{r^{n-1}}|}{h(r^{n-1})} \leq 1$ eventually. In this case, $\frac{|B_{r^{n-1}}|}{h(r^n)} < \frac{|B_{r^{n-1}}|}{\sqrt{r}h(r^{n-1})} < r^{-1/2}$ eventually. Thus,

$$\frac{B_{r^n}}{h(r^n)} \geq \frac{B_{r^{n-1}}}{h(r^n)} + \sqrt{\frac{r-1}{r}} \geq \sqrt{\frac{r-1}{r}} - r^{-1/2} = \frac{\sqrt{r-1}-1}{\sqrt{r}}$$

i.o. a.s.. Thus

$$\limsup_{t\to\infty} \frac{B_t}{h(t)} \geq \limsup_{n\to\infty} \frac{B_{r^n}}{h(r^n)} \geq \frac{\sqrt{r-1}-1}{\sqrt{r}} \quad \text{a.s.}$$

This is true for all $r > 1$. Taking a countable limit of the form $r \in \mathbb{Z}_{\geq 0}$, we conclude that $\limsup_{t\to\infty} \frac{B_t}{h(t)} \geq 1$ almost surely.

Problem 5.4 — What is the value of $\liminf_{t\to\infty} \frac{B_t}{h(t)}$?

By symmetry, it is -1 almost surely.